AST 6007-01 Spacecraft Dynamics and Control

Homework #4

Due Nov 17 2021, 10:00AM

1. A rigid spacecraft has principal moments of inertia (1,000, 1,400, 1,700) *kg-m*^{2}. This spacecraft is placed into a circular orbit at an altitude of $500km$ above the Earth.

   a. How should the three principal axes of the spacecraft be aligned with respect to LVLH coordinate frame so that the spacecraft is stable in pitch, roll and yaw according to the linearized equations for the rotational motion?

   b. Corresponding to the conditions you obtained in part a, what are the eigenvalues for the pitch dynamics and for the roll/yaw dynamics?

   c. Corresponding to the conditions you obtained in part a, simulate the nonlinear on-orbit spacecraft rotational equations to determine the Euler angle time responses, assuming that each initial Euler angle is 0.1 radians and the Euler angle rates are all initially zero. Compare these time responses with your results in part b.

2. A rigid spacecraft has principal moments of inertia $I_x = 1{,}200kg - m^2, I_y = 1{,}000kg - m^2, I_z = 800kg - m^2$, defined with respect to the spacecraft principal axes. This spacecraft is placed into a geo-synchronous circular orbit about the Earth so that its *x*-axis is aligned with the local horizontal direction in the orbital plane, its *y*-axis is aligned normal to the orbital plane, and its *z*-axis is aligned along the radial direction.

   a. What are the linearized pitch equations and their associated eigenvalues?

   b. What are the linearized roll/yaw equations and their associated eigenvalues?

   c. Simulate the nonlinear on-orbit spacecraft rotational equations to determine the Euler angle time responses, assuming that the initial Euler angles are $\phi = 0.1, \theta = 0.1, \psi = 0.0$ radians, and that the Euler angle rates are all initially zero. Compare these time responses with your results in part b.